

“Mirror, Mirror on the Wall: Which Jobs Will AI Replace After All?”:

A New Index of Occupational Exposure

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1 INTRODUCTION

The concern that more autonomous instruments and machines would render work obsolete has recurred throughout history. From Aristotle speculating on the obsolescence of human labor if instruments could work on their own to Keynes’ famous prediction of technological unemployment,² this concern arises whenever new technologies emerge and has particularly grown with recent advances in artificial intelligence (AI). Growing progress in natural language processing and generative models like OpenAI’s GPT have significantly improved machines’ abilities to mimic many human cognitive capabilities, such as understanding and generating human-like text, making predictions, and performing image and video recognition (Dwivedi *et al.*, 2023). Furthermore, the convergence of AI with robotics signals the emergence of highly autonomous systems capable of executing more complex physical tasks, potentially reshaping the landscape of both cognitive and manual labor.

This paper aims to identify which occupations are most exposed to the replacement due to AI adoption, developing a new Generated Index of Occupational Exposure (GENOE). Our approach uses a novel methodological approach that we call “synthetic AI surveys.” This methodology leverages the extensive knowledge and advanced analytical capabilities of large language models (LLMs) to conduct expert-like assessments, provide estimations, and offer insights on a wide range of areas. We specifically applied this approach to estimate the exposure of various tasks and occupations to recent and anticipated advancements in AI-related technologies. The GENOE index measures the likelihood of job replacement by AI at the occupational level, based on the tasks typically performed in each role. We defined three time horizons for replacement: one year, five years, and 10 years. This accounts for short-term scenarios based on current AI capabilities as well as medium-term and long-term scenarios anticipating future technological advancements.

Our findings reveal that the average occupational exposure to AI is 0.28 within one year, increasing to 0.38 and 0.44 over the five- and 10-year horizons, respectively.³ An exposure score of 0.28 suggests that, on average, occupations have a 28% likelihood of potentially being impacted by AI within the next year. To contextualize these results, we calibrated the GENOE index with labor market data from the United States and Mexico. Results show that 43 million and 16 million jobs are exposed to AI in the United States and Mexico, respectively, over the one-year horizon, increasing to 60 million and 22 million over five years, and 70 million and 26 million over ten years. Extrapolating these findings to Latin America and the Caribbean, we estimate that approximately 84 million jobs are exposed to AI within one year, rising to 114 million in 5 years, and 132 million in 10 years. Assuming this average exposure is applicable to worldwide labor markets, it implies that approximately 980 million, 1.33 billion, and 1.54 billion jobs are exposed to AI over the one-, five-, and 10-year horizons, respectively. It is crucial to note that this does not directly translate

²See Aristotle (350 B.C.E./1998), Politics and Keynes (1930).

³GENOE estimates are available online for use by academics, policymakers, and businesses to inform future research and strategic decision-making GENOE. Data can be found in this [online repository](#).